

Bayesian Linear Regression

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Linear Regression

- Model: $Y \approx \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3$
- Parameters: $\beta_0, \beta_1, \beta_2, \beta_3$
- Loss function

$$L(\beta_0, \beta_1, \beta_2, \beta_3) = \sum_i^N \left(y^{(i)} - \left(\beta_0 + \beta_1 x_1^{(i)} + \beta_2 x_2^{(i)} + \beta_3 x_3^{(i)} \right) \right)^2$$

- Task

$$\arg \min_{\beta_0, \beta_1, \beta_2, \beta_3} \sum_i^N \left(y^{(i)} - \left(\beta_0 + \beta_1 x_1^{(i)} + \beta_2 x_2^{(i)} + \beta_3 x_3^{(i)} \right) \right)^2$$

Linear Regression

- Model: $Y = \mathbf{w}^T \mathbf{x}$
- Parameters: $\mathbf{w} = [w_1, w_2, \dots, w_d]^T$
- Loss function

$$L(\mathbf{w}) = \sum_i^N \left(y^{(i)} - (\mathbf{w}^T \mathbf{x}) \right)^2$$

- Task

$$\arg \min_{\mathbf{w}} L(\mathbf{w})$$

Linear Regression

We can give linear regression a probabilistic interpretation by assuming a Gaussian noise model:

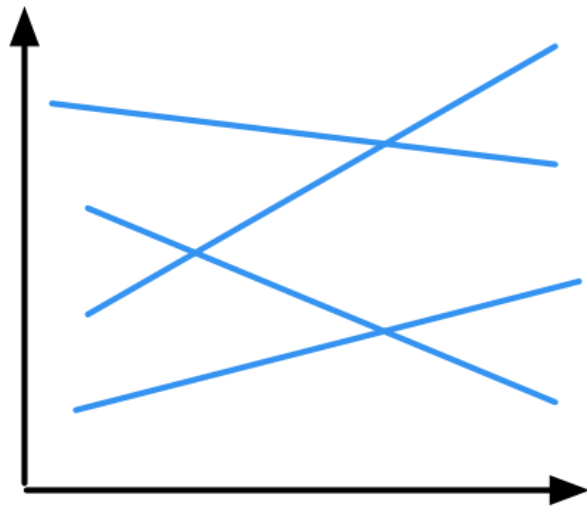
- Model: $Y = \mathbf{w}^T \mathbf{x} + \epsilon, \epsilon \sim N(0, \sigma^2) \Rightarrow y|\mathbf{x} \sim N(\mathbf{w}^T \mathbf{x}, \sigma^2)$
- Parameters: $\mathbf{w} = [w_1, w_2, \dots, w_d]^T$
- Log likelihood function

$$\begin{aligned} \frac{1}{N} \sum_i^N \log p(y^{(i)} | \mathbf{x}^{(i)}; \mathbf{w}) &= \frac{1}{N} \sum_i^N \log N(y^{(i)}; \mathbf{w}^T \mathbf{x}, \sigma^2) \\ &= -\frac{1}{2N\sigma^2} \sum_i^N (y^{(i)} - \mathbf{w}^T \mathbf{x})^2 + \text{const} \end{aligned}$$

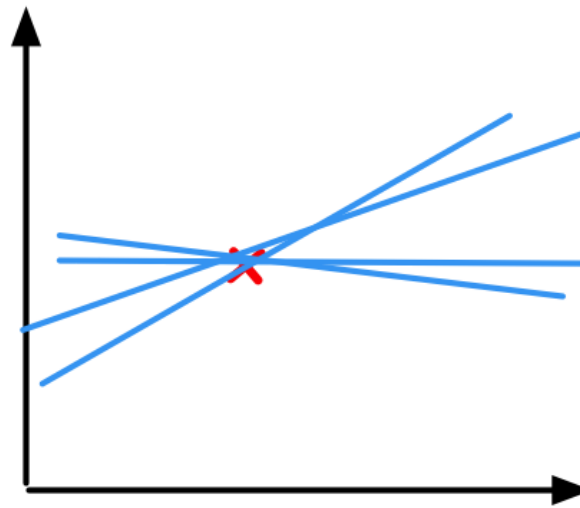
- Task: Maximize log likelihood function $\cong$ Minimize loss function

Bayesian Linear Regression

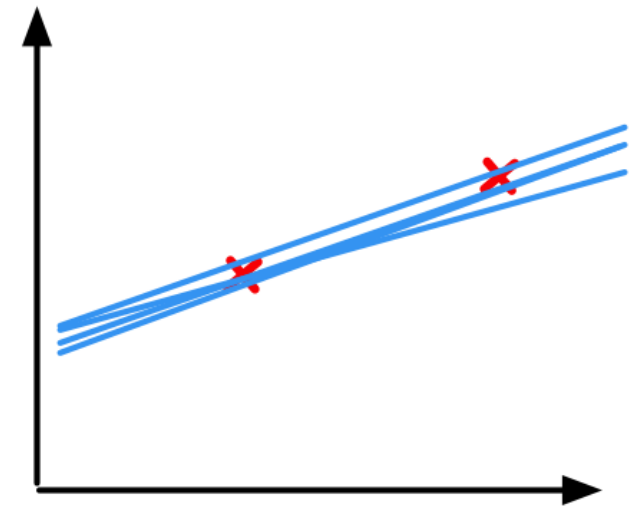
- Bayesian linear regression considers various plausible explanations for how the data were generated
- It makes predictions using all possible regression weights, weighted by their posterior probability



no observations



one observation



two observations

Gaussian Prior

We assume a Gaussian prior, $\mathbf{w} \sim N(\mathbf{m}, \mathbf{S})$

$$\begin{aligned}\log p(\mathbf{w}) &= \log N(\mathbf{w}; \mathbf{m}, \mathbf{S}) \\ &= -\frac{1}{2}(\mathbf{w} - \mathbf{m})^T \mathbf{S}^{-1}(\mathbf{w} - \mathbf{m}) + \textit{const}\end{aligned}$$

Bayesian Linear Regression

- Prior distribution: $\mathbf{w} \sim N(0, \mathbf{S})$
- Likelihood: $y|\mathbf{x}, \mathbf{w} \sim N(\mathbf{w}^T \mathbf{x}, \sigma^2)$
- Posterior distribution:

$$\begin{aligned}\log p(\mathbf{w}|D) &= \log p(\mathbf{w}) + \log p(D|\mathbf{w}) + \text{const} \\ &= -\frac{1}{2}(\mathbf{w} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{w} - \boldsymbol{\mu}) + \text{const}\end{aligned}$$

where $\boldsymbol{\mu} = \sigma^{-2} \boldsymbol{\Sigma} \mathbf{X}^T \mathbf{y}$, $\boldsymbol{\Sigma}^{-1} = \sigma^2 \mathbf{X}^T \mathbf{X} + \mathbf{S}^{-1}$

- So, posterior distribution: $\mathbf{w}|D \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$
- Since a Gaussian prior leads to a Gaussian posterior, the Gaussian distribution is the conjugate prior for linear regression

Bayesian Linear Regression

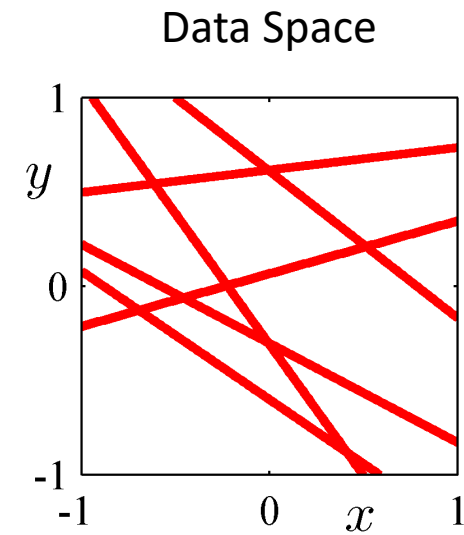
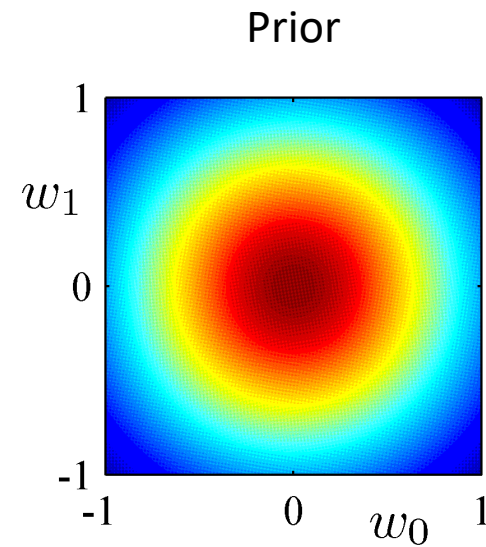
- Prior distribution: $\mathbf{w} \sim N(0, \mathbf{S})$
- Likelihood: $y|\mathbf{x}, \mathbf{w} \sim N(\mathbf{w}^T \mathbf{x}, \sigma^2)$
- Then, posterior distribution:

$$\begin{aligned}\log p(\mathbf{w}|D) &= \log p(\mathbf{w}) + \log p(D|\mathbf{w}) + \text{const} \\ &= -\frac{1}{2}(\mathbf{w} - 0)^T \mathbf{S}^{-1}(\mathbf{w} - 0) - \frac{1}{2N\sigma^2} \sum_i^N (y^{(i)} - \mathbf{w}^T \mathbf{x})^2 + \text{const} \\ &= -\frac{1}{2}(\mathbf{w})^T \mathbf{S}^{-1}(\mathbf{w}) - \frac{1}{2\sigma^2} \|\mathbf{y} - \mathbf{w}^T \mathbf{X}\|^2 + \text{const} \\ &= -\frac{1}{2}(\mathbf{w} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{w} - \boldsymbol{\mu}) + \text{const}\end{aligned}$$

where $\boldsymbol{\mu} = \sigma^{-2} \mathbf{\Sigma} \mathbf{X}^T \mathbf{y}$, $\boldsymbol{\Sigma}^{-1} = \sigma^2 \mathbf{X}^T \mathbf{X} + \mathbf{S}^{-1}$

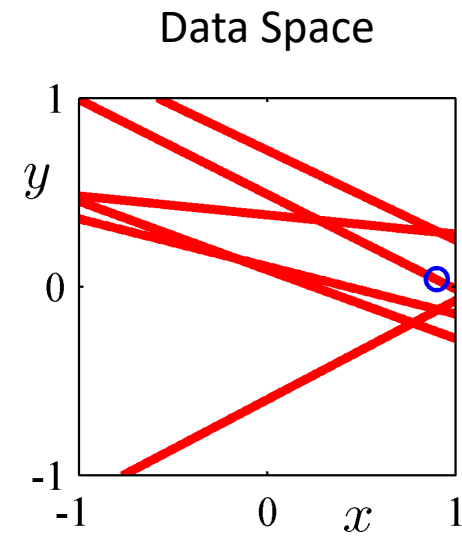
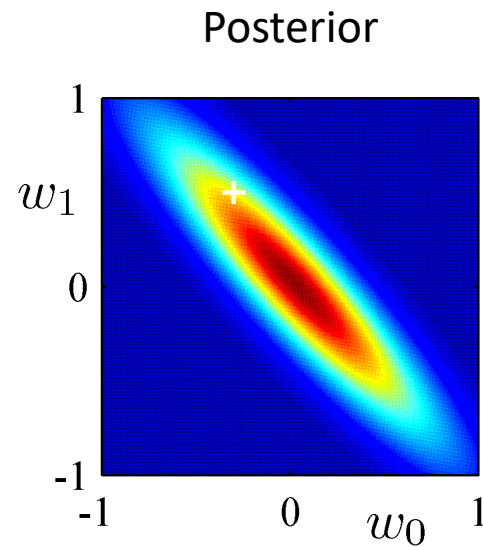
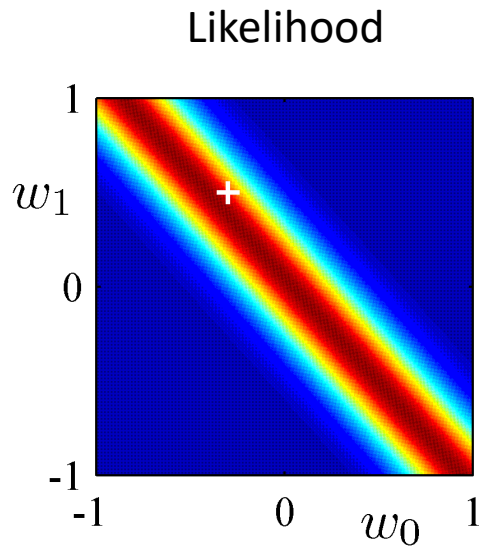
Bayesian Linear Regression

0 data points observed



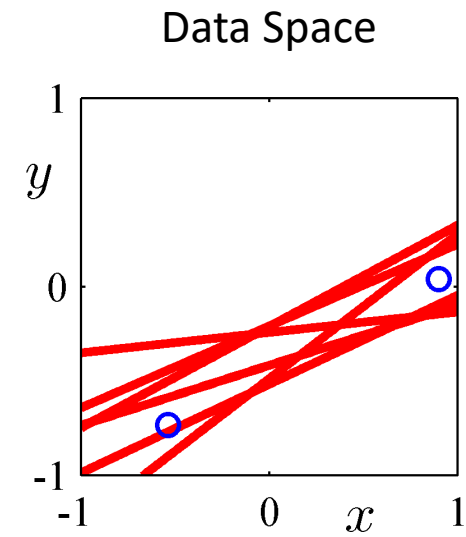
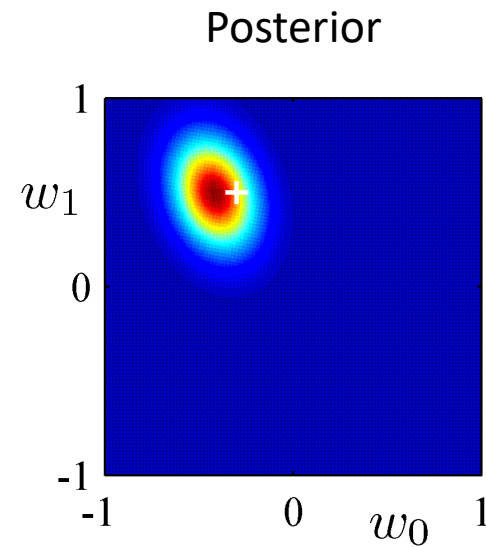
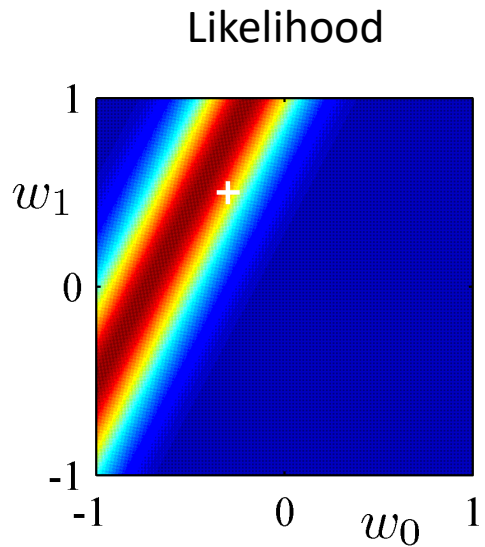
Bayesian Linear Regression

1 data point observed



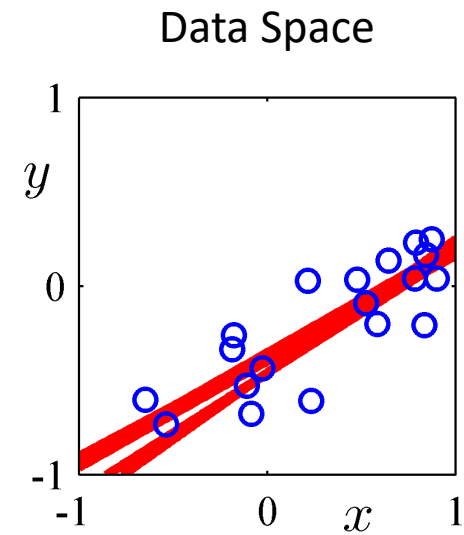
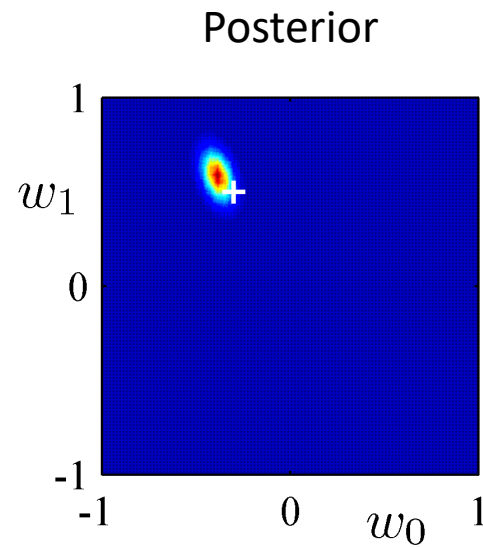
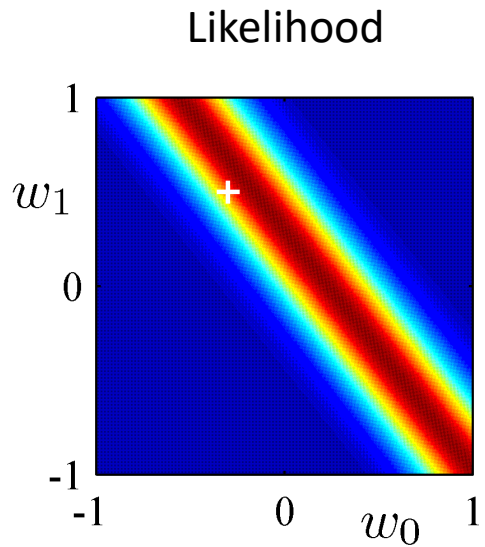
Bayesian Linear Regression

2 data points observed



Bayesian Linear Regression

20 data points observed



Predictive Distribution

- Predict y for new values of x by integrating over $\mathbf{w}$:

$$p(y|x, D) = \int \underbrace{p(y|x, \mathbf{w})}_{N(y; \mathbf{w}^T \mathbf{x}, \sigma^2)} \underbrace{p(\mathbf{w}|D)}_{N(\mathbf{w}; \boldsymbol{\mu}, \boldsymbol{\Sigma})} d\mathbf{w}$$

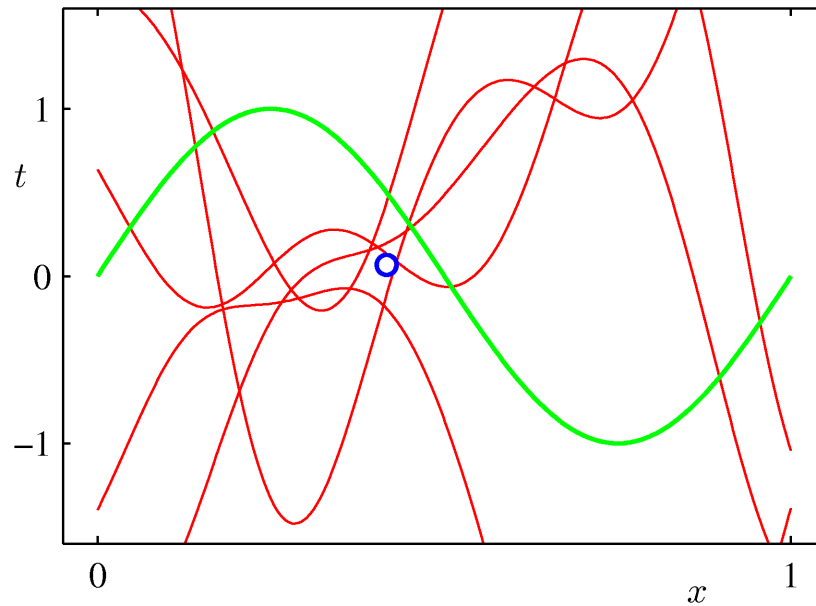
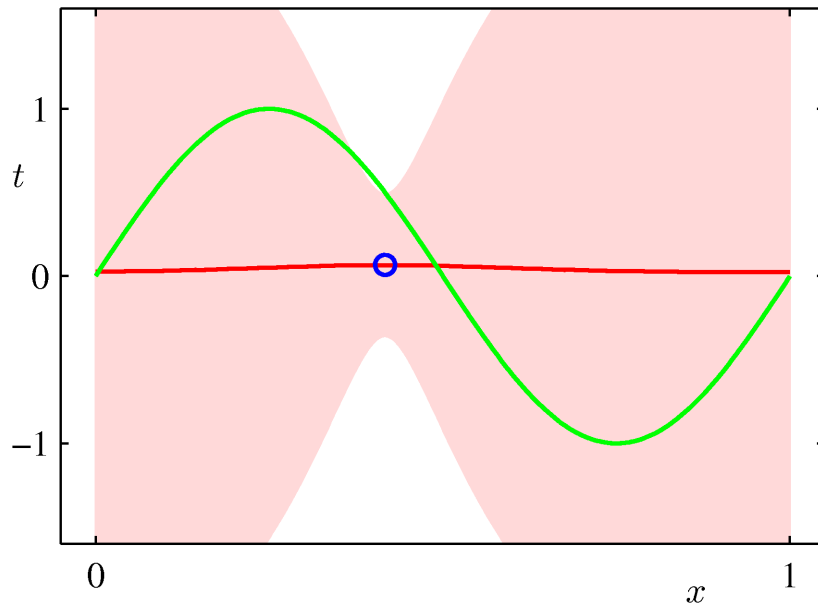
- By the linear combination rules for Gaussian random variables, y is a Gaussian distribution with parameters

$$\begin{aligned}\mu_{pred} &= \boldsymbol{\mu}^T \mathbf{x} \\ \sigma_{pred}^2 &= \mathbf{x}^T \boldsymbol{\Sigma} \mathbf{x} + \sigma^2\end{aligned}$$

- Hence, the posterior predictive distribution is $N(y; \mu_{pred}, \sigma_{pred}^2)$

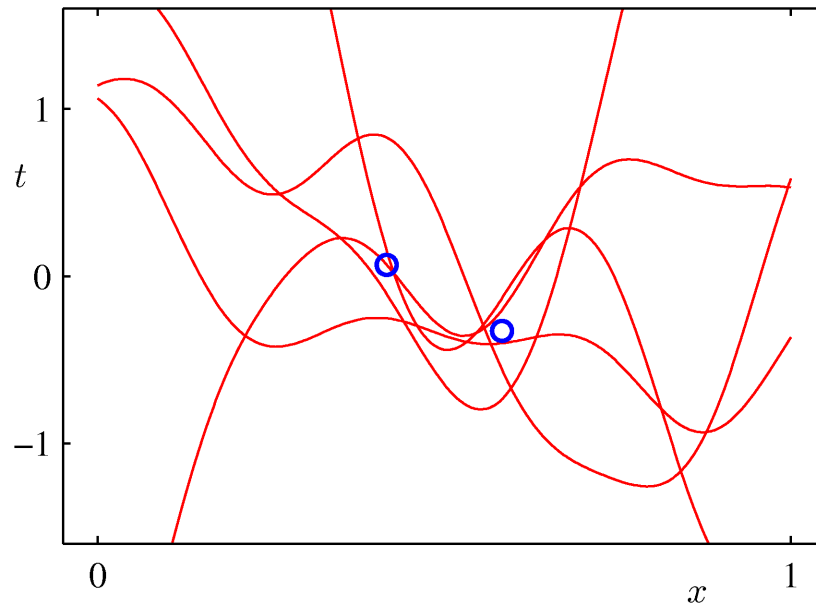
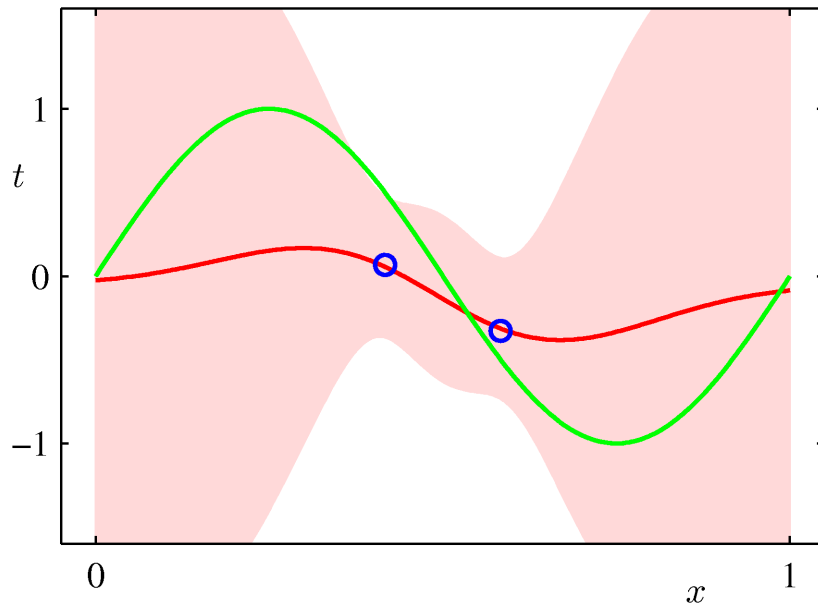
Predictive Distribution

Example: Sinusoidal data, 1 data point



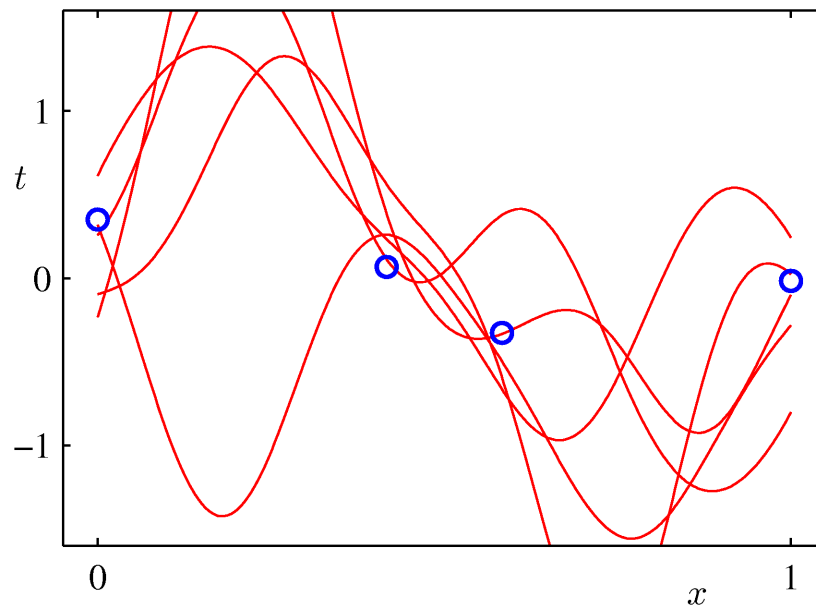
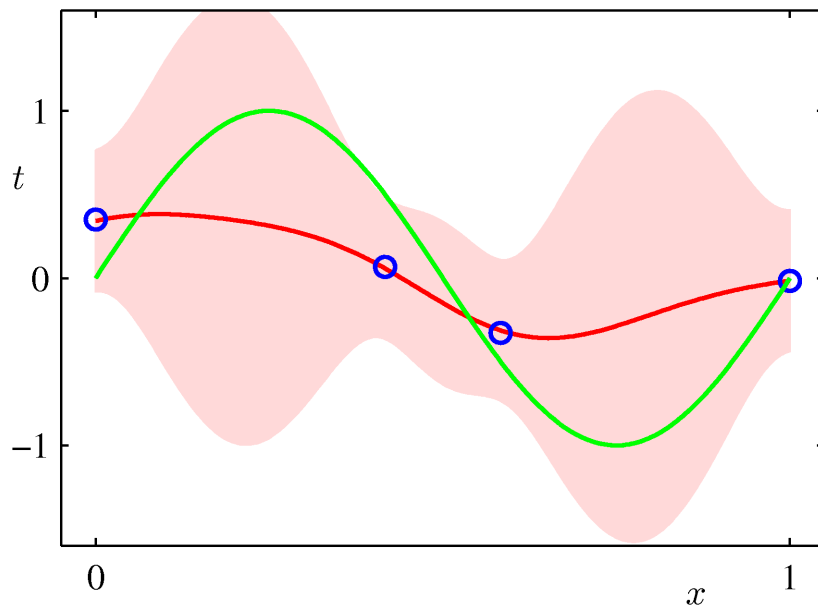
Predictive Distribution

Example: Sinusoidal data, 2 data points



Predictive Distribution

Example: Sinusoidal data, 4 data points



Predictive Distribution

Example: Sinusoidal data, 25 data points

